

Substance use disorder (SUD) is a complex condition in which there is uncontrolled use of a substance despite harmful consequences. People with SUD have an intense focus on using a certain substance(s) such as alcohol, tobacco, or illicit drugs, to the point where the person's ability to function in day-to-day life becomes impaired. People keep using the substance even when they know it is causing or will cause more harm than good.

SUDs are classified as mild, moderate, or severe based on how many of the 11 criteria are fulfilled: mild, any 2 or 3 criteria; moderate, any 4 or 5 criteria; severe, any 6 or more criteria.

Elevated risk of suicide

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**Pharmacologic  
(DSM-5 criteria 10 and  
11)**

- Tolerance:** Individual requires increasingly higher doses of the substance to achieve the desired effect, or the usual dose has a reduced effect; individuals may build tolerance to specific symptoms at different rates.
- Withdrawal:** A collection of signs and symptoms that occurs when blood and tissue levels of the substance decrease. Individuals are likely to seek the substance to relieve symptoms. No documented withdrawal symptoms from hallucinogens, PCP, or inhalants.
- Note:** Individuals can have an SUD with prescription medications, so tolerance and withdrawal (criteria 10 and 11) in the context of appropriate medical treatment do not count as criteria for an SUD.

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Alcohol

Unsteady gait, slurred  
speech, nystagmus,  
flushed face, conjunctival  
injection, decreased  
levels of consciousness

Disinhibition,  
argumentativeness,  
aggression, inattention,  
lability of mood, impaired  
judgement and functioning

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Alcohol poisoning can result from drinking any type of alcohol, including beer, wine or liquor. As your stomach digests and absorbs alcohol, the alcohol enters your bloodstream, and your alcohol blood level begins to rise. Your liver breaks down alcohol. But when blood alcohol levels are high, your overwhelmed liver can't remove the toxins quickly enough.

The extra alcohol in the bloodstream is a depressant. That means it reduces normal function. In this case, it affects the parts of the brain that control vital body functions, such as breathing, heart rate, blood pressure and temperature. As blood alcohol continues to rise, the depressant effect is more substantial

# Alcohol withdrawal syndrome

May occur within hours, peaks at 72 hours, results in the absence of GABAergic input of alcohol

May include tremors, insomnia, agitation, diaphoresis, hypertension, tachycardia or seizures

Hallucinations may occur and signal alcohol withdrawal DELIRIUM (2% of patients)

Any history of previous withdrawal from alcohol that includes delirium or seizures should be treated in a monitored setting

# Nicotine withdrawal

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- The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) reports 7 primary symptoms associated with nicotine withdrawal: irritability/anger/frustration, anxiety, depressed mood, difficulty concentrating, increased appetite, insomnia, and restlessness
- There are seven FDA-approved medications for smoking cessation, including two non-nicotine replacement pill options of bupropion (Zyban) and varenicline (Chantix) and five nicotine-replacement therapies (NRTs).
- Varenicline (Chantix) appears to be the most effective compared with placebo, followed by bupropion and NRTs. Varenicline is a partial agonist at the  $\alpha 4\beta 2$  receptors; as a partial agonist, it relieves craving and withdrawal, but reduces the reinforcing effects of nicotine by blocking dopaminergic stimulation (less smoking reinforcement/reward).
- Bupropion (Zyban) is the other medication that can affect dopamine, norepinephrine, and nicotinic-cholinergic receptors to decrease cravings and withdrawal symptoms.
- NRTs function to satisfy nicotine receptors while being less reinforcing.

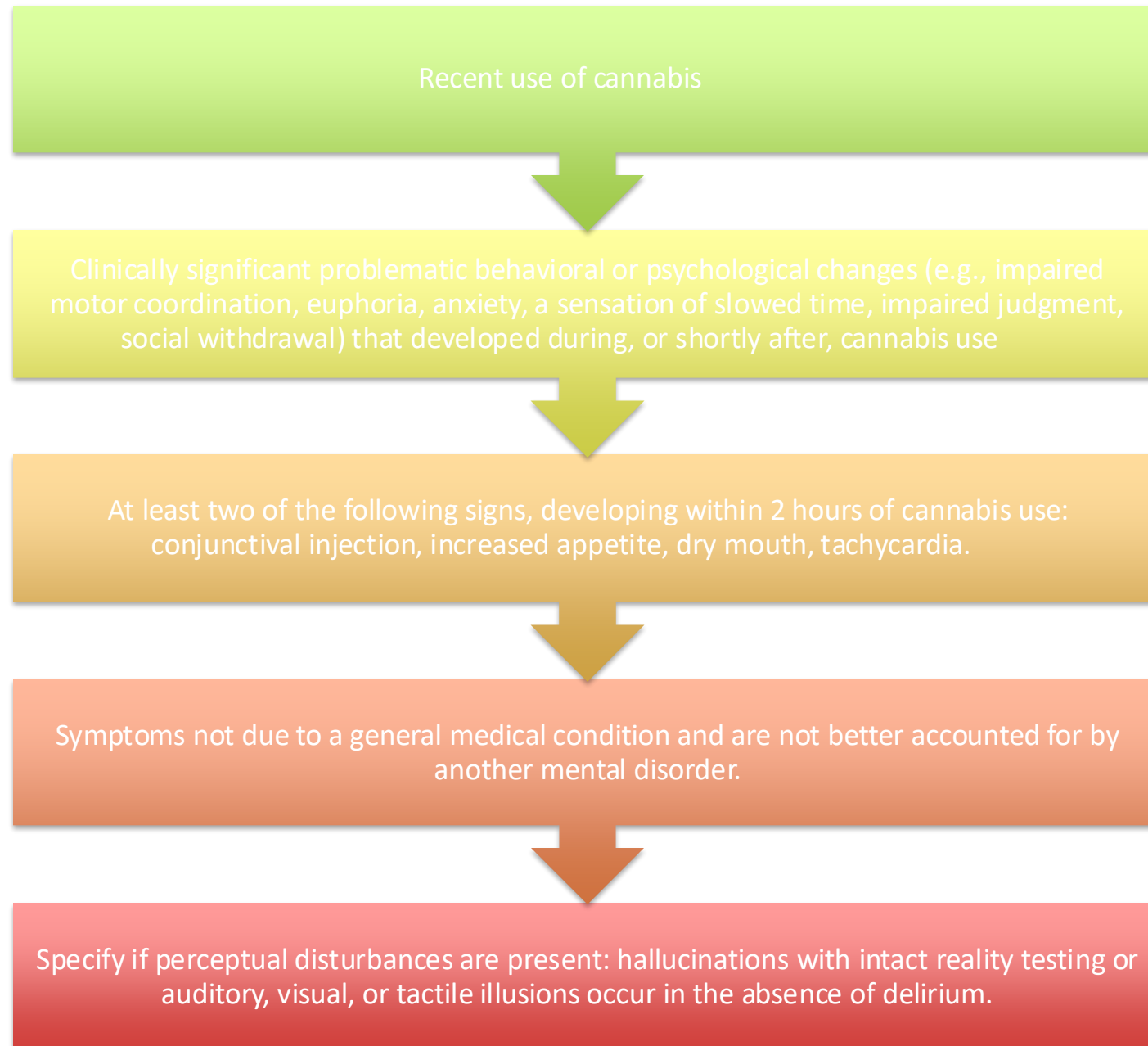


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# Cannabis use disorder

- The acute phase includes intoxication and withdrawal states, along with secondary complications - delirium, psychosis, anxiety, and insomnia. Chronic regular use can be characterized by disordered behavior (DMS 5)



This accompanies cessation of cannabis use that has been heavy and prolonged (i.e., usually daily or almost daily use over a period of at least a few months). Three or more of the following signs and symptoms develop within approximately 1 week after cessation of heavy, prolonged use:

- Irritability, anger, or aggression
- Nervousness or anxiety
- Sleep difficulty (i.e., insomnia, disturbing dreams)
- Decreased appetite or weight loss
- Restlessness
- Depressed mood

At least one of the following physical symptoms causing significant discomfort: abdominal pain, shakiness/tremors, sweating, fever, chills, or a headache.

The signs or symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

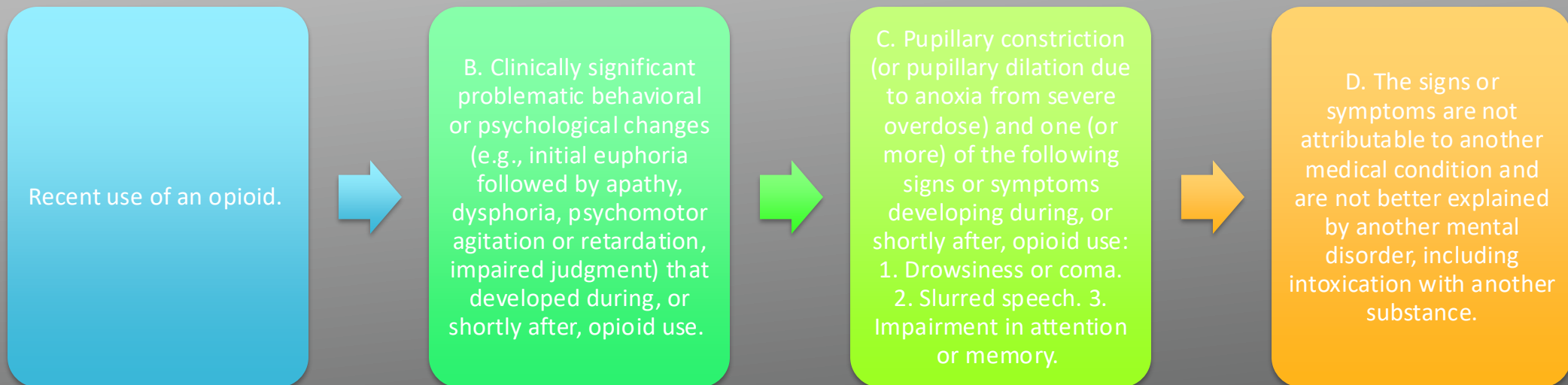
The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

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A. Presence of either of the following: 1. Cessation of (or reduction in) opioid use that has been heavy and prolonged (i.e., several weeks or longer). 2. Administration of an opioid antagonist after a period of opioid use.

B. Three (or more) of the following developing within minutes to several days after Criterion A: 1. Dysphoric mood. Nausea or vomiting. Muscle aches. Lacrimation or rhinorrhea. Pupillary dilation, piloerection, or sweating. Diarrhea. Yawning. Fever. Insomnia.

C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

# Treatment of Opioid Use Disorder

Methadone is an opioid **agonist**/eliminates withdrawal and reduces cravings

Buprenorphine is a **partial opioid agonist**/reduces withdrawal and cravings

Naltrexone is an **opioid antagonist**/prevent opioids from producing rewarding effects

Methadone=Buprenorphine>naltrexone

All are more effective with counseling/behavioral therapy

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A. Recent use of phencyclidine (or a pharmacologically similar substance).



B. Clinically significant problematic behavioral changes (e.g., belligerence, assaultiveness, impulsiveness, unpredictability, psychomotor agitation, impaired judgment) that developed during, or shortly after, phencyclidine use.



C. Within 1 hour, two (or more) of the following signs or symptoms: Note: When the drug is smoked, "snorted," or used intravenously, the onset may be particularly rapid.  
1. Vertical or horizontal nystagmus. 2. Hypertension or tachycardia. Numbness or diminished responsiveness to pain. Ataxia. Dysarthria. Muscle rigidity. Seizures or coma. Hyperacusis.



D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance

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Recent intended or unintended short-term, high-dose exposure to inhalant substances, including volatile hydrocarbons such as toluene or gasoline.



B. Clinically significant problematic behavioral or psychological changes (e.g., belligerence, assaultiveness, apathy, impaired judgment) that developed during, or shortly after, exposure to inhalants.



C. Two (or more) of the following signs or symptoms developing during, or shortly after, inhalant use or exposure: Dizziness. Nystagmus. Incoordination. Slurred speech. Unsteady gait. Lethargy. Depressed reflexes. Psychomotor retardation. Tremor. Generalized muscle weakness. Blurred vision or diplopia. Stupor or coma. Euphoria.



D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

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A. Recent use of a sedative, hypnotic, or anxiolytic.

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graph TD; A[A. Recent use of a sedative, hypnotic, or anxiolytic.] --> B[B. Clinically significant maladaptive behavioral or psychological changes (e.g., inappropriate sexual or aggressive behavior, mood lability, impaired judgment) that developed during, or shortly after, sedative, hypnotic, or anxiolytic use.]; B --> C[C. One (or more) of the following signs or symptoms developing during, or shortly after, sedative, hypnotic, or anxiolytic use: Slurred speech. Incoordination. Unsteady gait. Nystagmus. Impairment in cognition (e.g., attention, memory). Stupor or coma.]; C --> D[D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.]; D --> E[Differentiate from alcohol intoxication];
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B. Clinically significant maladaptive behavioral or psychological changes (e.g., inappropriate sexual or aggressive behavior, mood lability, impaired judgment) that developed during, or shortly after, sedative, hypnotic, or anxiolytic use.

C. One (or more) of the following signs or symptoms developing during, or shortly after, sedative, hypnotic, or anxiolytic use: Slurred speech. Incoordination. Unsteady gait. Nystagmus. Impairment in cognition (e.g., attention, memory). Stupor or coma.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Differentiate from alcohol intoxication

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A. Cessation of (or reduction in) sedative, hypnotic, or anxiolytic use that has been prolonged.



B. Two (or more) of the following, developing within several hours to a few days after the cessation of (or reduction in) sedative, hypnotic, or anxiolytic use described in Criterion A: 1. Autonomic hyperactivity (e.g., sweating or pulse rate greater than 100 bpm). 2. Hand tremor. Insomnia. . Nausea or vomiting. . Transient visual, tactile, or auditory hallucinations or illusions. . Psychomotor agitation. Anxiety. . Grand mal seizures.



C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.



D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

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A. Recent use of an amphetamine-type substance, cocaine, or other stimulant.

B. Clinically significant problematic behavioral or psychological changes (e.g., euphoria or affective blunting; changes in sociability; hypervigilance; interpersonal sensitivity; anxiety, tension, or anger; stereotyped behaviors; impaired judgment) that developed during, or shortly after, use of a stimulant.

C. Two (or more) of the following signs or symptoms, developing during, or shortly after, stimulant use: Tachycardia or bradycardia. Pupillary dilation. Elevated or lowered blood pressure. Perspiration or chills. Nausea or vomiting. Evidence of weight loss. Psychomotor agitation or retardation. Muscular weakness, respiratory depression, chest pain, or cardiac arrhythmias. Confusion, seizures, dyskinesias, dystonias, or coma.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Chest pain, palpitations are a frequent presenting symptom.

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A. Cessation of (or reduction in) prolonged amphetamine-type substance, cocaine, or other stimulant use.



B. Dysphoric mood and two (or more) of the following physiological changes, developing within a few hours to several days after Criterion A: Fatigue. Vivid, unpleasant dreams. Insomnia or hypersomnia. Increased appetite. Psychomotor retardation or agitation.



C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.



D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.